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DOCUMENT CC 101 Laboratory Activity: Exploring the Windows Command Line

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BAGONG PILIPINAS

CC 101 Laboratory Activity: Exploring the Windows Command Line

Item	Details
Course	CC 101 Introduction to Computing (BSCS 1st Year)
Course Outcome	CO2: Identify the functions and relationships of major hardware and software components
Output	Command Line Exploration Documentation (PDF)
Estimated time	2 to 2½ hours
Reference	CC 101 Lecture Notes: Exploring the Windows Command Line

Instructions

Follow every step in order, take the screenshots marked **[Screenshot]**, and answer the questions marked **[Answer]** in your documentation.

Before you start

- You need a Windows 10 or 11 computer. No administrator rights are needed.
- To take a screenshot, press **Windows + Shift + S**, drag over the terminal window, then paste it (Ctrl + V) into your documentation.
- Replace the sample values (Juan Dela Cruz, BSCS 1-1) with **your own** details.
- Type every command yourself. Do not copy and paste; typing is how you learn.

Part A: Open the Two Shells

1. Press Windows + R, type cmd, and press Enter. Type cls and press Enter.
2. Type echo Juan Dela Cruz, BSCS 1-1 (your own name and section) and press Enter. **[Screenshot] Figure A1**

```
C:\Users\Billy>echo Billy Jhoe Gacias, BSCS 1-1
Billy Jhoe Gacias, BSCS 1-1
```

Figure A1

3. Press Windows + R, type powershell, and press Enter. Type the same echo command, but put the text in quotation marks: echo "Juan Dela Cruz, BSCS 1-1". PowerShell treats the comma as a list separator without them. **[Screenshot] Figure A2**

```
PS C:\Users\Billy> echo "Billy Jhoe Gacias, BSCS 1-1"
Billy Jhoe Gacias, BSCS 1-1
```

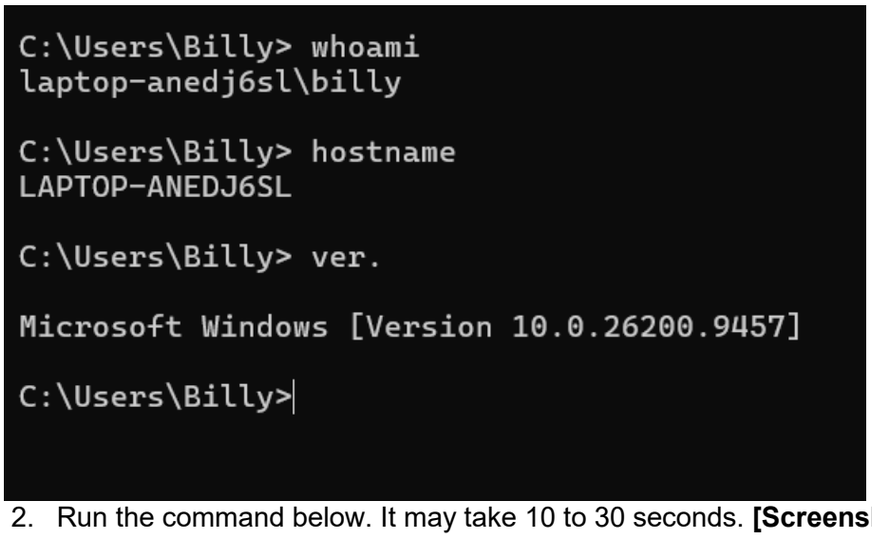
Figure A2

[Answer] A-Q1. Compare the two prompts. What is different, and what does each part of the Command Prompt prompt mean? *In Figure A1 it shows that it does not need to have quotation marks to function, now if I type it like this on the PowerShell it will result in a different output. Although if I put quotation on cmd it will also result in having the quotation in it. Albeit in Figure A2 it will function as shown in Figure A1.*

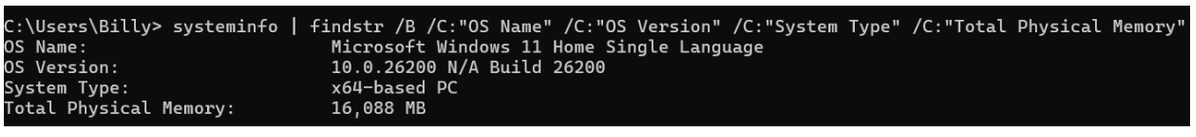
Part B: Inspect Your System (Command Prompt)

Use the Command Prompt window for Parts B to D.

1. Run these commands one at a time: whoami, then hostname, then ver. **[Screenshot] Figure B1**



2. Run the command below. It may take 10 to 30 seconds. **[Screenshot] Figure B2**




```
systeminfo | findstr /B /C:"OS Name" /C:"OS Version" /C:"System Type" /C:"Total
Physical Memory"
```

3. Run ipconfig. Find the adapter that shows an IPv4 Address. **[Screenshot] Figure B3**



[Answer] B-Q1. Complete this table in your documentation using your results:

Item	Your result
Logged-in account (whoami)	laptop-anedj6sl\billy
Computer name (hostname)	LAPTOP-ANEDJ6SL
OS name	Microsoft Windows 11 Home Single Language
OS version	10.0.26200 N/A Build 26200
System type (x64 or ARM)	x64-based PC
Total physical memory (RAM)	16,088 MB
IPv4 address	192.168.0.114

[Answer] B-Q2. The | symbol in step 2 is called a **pipe**. Run systeminfo alone and compare. In your own words, what did | findstr do? | findstr finds the specific item or variable that was typed, getting the exact things rather than showing everything that systeminfo shows.

Part C: Build a Folder Structure

1. Go to your user folder and create the lab folders:

```
cd %USERPROFILE%
mkdir CC101-Lab
cd CC101-Lab
mkdir Notes Projects Screenshots
```

2. Create a text file about yourself (use your own details):

```
echo Name: Juan Dela Cruz > profile.txt
echo Section: BSCS 1-1 >> profile.txt
echo Favorite app: YouTube >> profile.txt
type profile.txt
```

```
C:\Users>cd Billy

C:\Users\Billy>Mkdir CC101-Lab

C:\Users\Billy>cd CC101-Lab

C:\Users\Billy\CC101-Lab>mkdir Notes Projects Screenshots
```

```
C:\Users\Billy\CC101-Lab> echo Name: Billy Jhoe F. Gacias > profile.txt

C:\Users\Billy\CC101-Lab> echo Section: BSCS 1-1 >> profile.txt

C:\Users\Billy\CC101-Lab> echo Favorite app: Krita >> profile.txt

C:\Users\Billy\CC101-Lab> type profile.txt
Name: Billy Jhoe F. Gacias
Section: BSCS 1-1
Favorite app: Krita
```

Figure C1

[Screenshot] Figure C1 (show all commands and the output of type)

3. Run tree /F to display your folder structure.

```
C:\Users\Billy\CC101-Lab> tree /F
Folder PATH listing for volume Acer
Volume serial number is 029B-3B52
C:.
    profile.txt

├───Notes
├───Projects
└───Screenshots
```

Figure C2

[Screenshot] Figure C2

4. Open File Explorer, go to C:\Users\(your account)\CC101-Lab, and open profile.txt in Notepad.

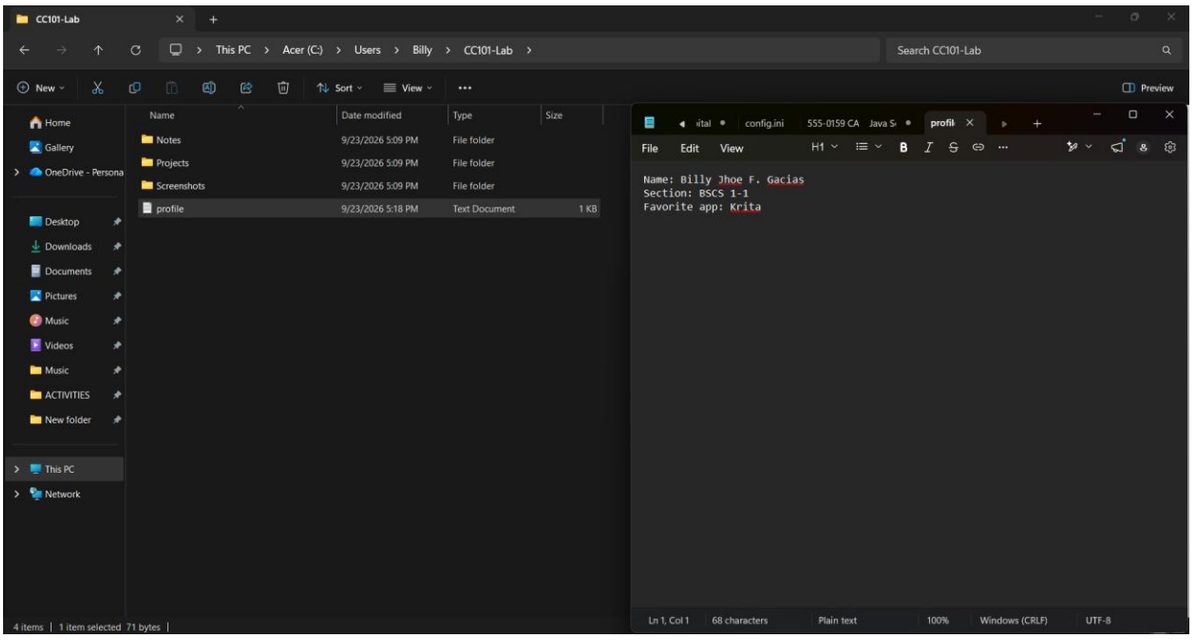


Figure C3

[Screenshot] Figure C3 (File Explorer and Notepad side by side)

[Answer] C-Q1. What would happen if you used > instead of >> in the second echo line? Test it on a new file (test.txt) and explain it. *It doesn't show the first thing you made as a profile.txt |That's why it's not appearing. It is disregarding the first word in the note that should be made.*

Part D: Copy, Rename, and Delete Files

```
copy profile.txt Notes\profile-copy.txt
cd Notes
ren profile-copy.txt about-me.txt
echo This file will be deleted > temp.txt
dir
del temp.txt
dir
cd ..
tree /F
```

```
C:\Users\Billy\CC101-Lab>copy profile.txt Notes\profile-copy.txt
        1 file(s) copied.

C:\Users\Billy\CC101-Lab>cd Notes

C:\Users\Billy\CC101-Lab\Notes>ren profile-copy.txt about-me.txt

C:\Users\Billy\CC101-Lab\Notes>echo This file will be deleted > temp.txt

C:\Users\Billy\CC101-Lab\Notes>dir
 Volume in drive C is Acer
 Volume Serial Number is 029B-3B52

 Directory of C:\Users\Billy\CC101-Lab\Notes

09/23/2026  05:27 PM    <DIR>          .
09/23/2026  05:23 PM    <DIR>          ..
09/23/2026  05:18 PM                71 about-me.txt
09/23/2026  05:27 PM                28 temp.txt
               2 File(s)             99 bytes
               2 Dir(s)  155,676,291,072 bytes free
```

Figure D1

```
C:\Users\Billy\CC101-Lab\Notes>del temp.txt

C:\Users\Billy\CC101-Lab\Notes>dir
 Volume in drive C is Acer
 Volume Serial Number is 029B-3B52

 Directory of C:\Users\Billy\CC101-Lab\Notes

09/23/2026  05:27 PM    <DIR>          .
09/23/2026  05:23 PM    <DIR>          ..
09/23/2026  05:18 PM                71 about-me.txt
               1 File(s)             71 bytes
               2 Dir(s)  155,675,426,816 bytes free

C:\Users\Billy\CC101-Lab\Notes>cd ..

C:\Users\Billy\CC101-Lab>tree/F
Folder PATH listing for volume Acer
Volume serial number is 029B-3B52
C:.
|   profile.txt
|   test.txt
|
+---Notes
|       about-me.txt
|
+---Projects
\---Screenshots
```

Figure D2

[Screenshot] Figure D1 (from copy to the first dir) and **[Screenshot] Figure D2** (from del to tree /F)

[Answer] D-Q1. After del temp.txt, check your Recycle Bin. Is temp.txt there? What does this tell you about deleting files from the command line? *The temp.txt wasn't there surprisingly, I assume that it removes it automatically from the Recycle Bin, I learnt that everything does not permanently get removed. It simply scrambles the binary, I assume it's still there just invisible.*

Part E: Explore PowerShell

Switch to the PowerShell window.

1. Go to your lab folder and list and read its files:

```
cd $HOME\CC101-Lab
Get-ChildItem
Get-Content profile.txt
$PSVersionTable
```

```
PS C:\Users\Billy> cd CC101-Lab
PS C:\Users\Billy\CC101-Lab> cd $HOME\CC101-Lab
PS C:\Users\Billy\CC101-Lab> Get-ChildItem


    Directory: C:\Users\Billy\CC101-Lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         9/23/2026   5:27 PM                Notes
d-----         9/23/2026   5:09 PM                Projects
d-----         9/23/2026   5:09 PM                Screenshots
-a----         9/23/2026   5:18 PM             71 profile.txt
-a----         9/23/2026   5:23 PM             20 test.txt


PS C:\Users\Billy\CC101-Lab> Get-Content profile.txt
Name: Billy Jhoe F. Gacias
Section: BSCS 1-1
Favorite app: Krita
PS C:\Users\Billy\CC101-Lab> $PSVersionTable

Name                           Value
----                           -----
PSVersion                      5.1.26100.9444
PSEdition                      Desktop
PSCompatibleVersions           {1.0, 2.0, 3.0, 4.0...}
BuildVersion                   10.0.26100.9444
CLRVersion                     4.0.30319.42000
WSManStackVersion              3.0
PSRemotingProtocolVersion      2.3
SerializationVersion           1.1.0.1
```

Figure E1

[Screenshot] Figure E1

2. Run Get-Help Get-ChildItem. If PowerShell asks whether to run Update-Help, type N and press Enter.

```
PS C:\Users\Billy\CC101-Lab> Get-Help Get-ChildItem

NAME
    Get-ChildItem

SYNTAX
    Get-ChildItem [[-Path] <string[]>] [[-Filter] <string>] [-Include <string[]>] [-Exclude <string[]>] [-Recurse] [-Depth <uint32>] [-Force] [-Name]
    [-UseTransaction] [-Attributes {ReadOnly | Hidden | System | Directory | Archive | Device | Normal | Temporary | SparseFile | ReparsePoint |
    Compressed | Offline | NotContentIndexed | Encrypted | IntegrityStream | NoScrubData}] [-FollowSymlink] [-Directory] [-File] [-Hidden] [-ReadOnly]
    [-System]  [<CommonParameters>]

    Get-ChildItem [[-Filter] <string>] -LiteralPath <string[]> [-Include <string[]>] [-Exclude <string[]>] [-Recurse] [-Depth <uint32>] [-Force] [-Name]
    [-UseTransaction] [-Attributes {ReadOnly | Hidden | System | Directory | Archive | Device | Normal | Temporary | SparseFile | ReparsePoint |
    Compressed | Offline | NotContentIndexed | Encrypted | IntegrityStream | NoScrubData}] [-FollowSymlink] [-Directory] [-File] [-Hidden] [-ReadOnly]
    [-System]  [<CommonParameters>]


ALIASES
    gci
    ls
    dir


REMARKS
    Get-Help cannot find the Help files for this cmdlet on this computer. It is displaying only partial help.
        -- To download and install Help files for the module that includes this cmdlet, use Update-Help.
        -- To view the Help topic for this cmdlet online, type: "Get-Help Get-ChildItem -Online" or
           go to https://go.microsoft.com/fwlink/?LinkID=113308.
```

Figure E2

[Screenshot] Figure E2

3. Take the **last three digits of your student ID number** (for example, 2025-00347 → 347). Convert it with PowerShell:

```
[Convert]::ToString(347, 2)
[Convert]::ToString(347, 8)
'{0:X}' -f 347
```

```
PS C:\Users\Billy\CC101-Lab> [Convert]::ToString(918, 2)
1110010110
PS C:\Users\Billy\CC101-Lab> [Convert]::ToString(918, 8)
1626
PS C:\Users\Billy\CC101-Lab> '{0:X}' -f 918
396
```

Figure E3

[Screenshot] Figure E3

[Answer] E-Q1. Write your number in decimal, binary, octal, and hexadecimal. Keep this; you will verify it by hand in Week 5. 1110010110, 1626, 396

Part F: Discover a New Command

1. In Command Prompt, type help to see a list of commands.
2. Choose **one** command that was **not** taught in this lesson and is safe to run (for example: date /t, time /t, tasklist, vol, or color 0A).
3. Read its help with command /? (for example, tasklist /?), then run it.

```
C:\Users\Billy\CC101-Lab>date /t
Wed 09/23/2026
```

Figure F1

```
C:\Users\Billy\CC101-Lab>date /?
Displays or sets the date.

DATE [/T | date]

Type DATE without parameters to display the current date setting and
a prompt for a new one.  Press ENTER to keep the same date.

If Command Extensions are enabled the DATE command supports
the /T switch which tells the command to just output the
current date, without prompting for a new date.
```

Figure F2

[Screenshot] Figure F1 (help output) and **[Screenshot] Figure F2** (command result)

[Answer] F-Q1. What does your chosen command do, and when would it be useful? *The command that I chose was the date, it shows the current date that is integrated to the laptop. It would be useful to know the date and year or what day it is currently.*

Part G: Reflection

[Answer] Answer in 3 to 5 sentences each:

1. **G-Q1.** Which task in this activity was faster in the CLI than in the GUI? Which was easier in the GUI? Explain why. *CLI was generally easier in programming folders and text without the use of GUI, although, I have been used to using GUI since it's more familiar to me, therefore it seems like it's easier to use them both. However, using CLI for more complex usage is more prominent to use because you only need to type to find anything that has been lost in your computer that GUI cannot see.*
2. **G-Q2.** What error did you encounter (if any), what caused it, and how did you fix it? *My errors were basically having wrong spelling or anything that isn't existing and spacing, it's hard to copy the exact prompts however with practice it should become much easier., while seeing the problem I fixed it when it shows what the problem was.*
3. **G-Q3.** Give one real-world job or situation in computing where knowing the command line would be important. *Some files or servers are not shown in GUI, so using command would be helpful finding these files that GUI can't access or see.*

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Documentation and Submission

Submit one PDF that records what you did, what the computer showed, and what you learned. Someone reading it should be able to repeat your work.

Required Contents (in this order)

1. **Cover page:** activity title, your full name, student ID, section, date performed, and instructor's name.
2. **Introduction** (3 to 5 sentences): what the command line is and the purpose of this activity, in your own words.
3. **Procedure and results, Parts A to F:** for each part, a short heading, the commands you typed, the labeled screenshots, and your answers to that part's [Answer] questions.
4. **Commands learned table:** every command you used and what it does, in your own words.
5. **Reflection:** your answers to G-Q1 to G-Q3.

Formatting Rules

Item	Requirement
File type	PDF (write in Word or Google Docs, then export as PDF)
File name	CC101_CLI-Lab_Lastname_Firstname.pdf
Font	Arial or Calibri, 11 pt; commands in Consolas or Courier New
Screenshots	Crop to the terminal window; text must be readable. Label each one below the image, e.g., "Figure B2. Filtered systeminfo output"
Personal details	Your own name, section, and student ID must appear in Figures A1, C1, and E3
Total screenshots	16 (A1–A2, B1–B3, C1–C3, D1–D2, E1–E3, F1–F2)

Sample Commands Learned Table

Command	Shell	What it does
mkdir Notes	Command Prompt	Creates a folder named Notes in the current folder
type profile.txt	Command Prompt	Displays the contents of profile.txt

Submission

- Upload the PDF to the activity page in the course LMS before the deadline your instructor sets.
- Outputs must be your own work. Identical screenshots or answers will be treated as plagiarism under the class policy.
- Keep your CC101-Lab folder; later activities may build on it.

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Grading Rubric

The documentation is scored out of 100 points and counts under Laboratory: Project.

Criterion	Points	Full marks	Partial marks	Low marks
Completeness of procedure	30	All 16 labeled, readable screenshots; every part A to F documented	1 to 4 screenshots missing or unreadable	5 or more screenshots missing
Accuracy of results	20	System-information table complete; commands and conversions correct and personalized	Minor errors or 1 to 2 missing values	Many errors, or results not the student's own
Answers to guide questions (A to F)	20	Correct, clear, in the student's own words	Mostly correct but brief or vague	Missing, incorrect, or copied
Reflection (G-Q1 to G-Q3)	15	Specific, honest, and linked to the activity	General statements with few specifics	Missing or off-topic
Organization and formatting	15	Follows the required order, naming, and formatting rules; commands learned table complete	Some rules not followed	Disorganized; wrong file type or name
Total	**100**			